

MasterControl

Business Problem:

MasterControl is a software service provider with two products, QX (Quality Solutions) and MX (Manufacturing solutions). MX is a much less established product, having launched about 4 years ago, compared to QX, which has been in the market for over 20 years. MX currently underperforms QX with a sales conversion rate of 12.7% versus 19.7% for QX. MasterControl believes the current targeting strategy is resulting in missed opportunities and wasted resources prospecting the wrong kind of leads.

Benefit of Solution:

By understanding the companies and job titles MasterControl should target for the MX product, the firm can more effectively allocate and focus its sales resources, thereby increasing conversion rates and ultimately expanding market share and revenue. Increasing the MX conversion rate to match that of QX would represent an approximate 50% improvement in sales success for the MX product.

Analytics Approach:

This analysis will develop an interpretable lead scoring framework that combines stakeholder profiling with individual lead prioritization. Through feature engineering and descriptive analysis of historical data, we will identify which contact type profiles (job title patterns combined with account characteristics) exhibit highest Mx conversion rates. We will then compare multiple classification models (logistic regression, random forest, gradient boosting) using ROC-AUC as the primary evaluation metric to select the model that best discriminates between high and low-potential leads. The selected model will be used to generate probability scores for individual leads based on their match to high-converting profiles, producing a ranked prioritization list for sales. Feature importance analysis will ensure the model's logic is interpretable and actionable. Success will be measured by the conversion lift achieved when sales focuses on top-scored leads, with the goal of increasing Mx conversion from the current 12.7% baseline toward the Qx benchmark of 19.7%.

Scope:

The deliverable for this project will be a list of industries and prospective job titles for sales representatives at MasterControl to target for lead prospecting. Our findings will be job titles and industries/markets that are more likely to convert to purchasing the Mx product, with the goal of increasing the conversion rate. Classifying which customer types are likely to churn, or predicting the churn rate, is outside the scope of this engagement. The sole purpose of this project is target marketing for MasterControl sales representatives, helping them determine which prospects are likely to convert and the decision-maker within each target company. Additionally, recommendations for improving the prospect information on the MasterControl homepage will be provided.

Success Metrics:

The primary success metric for this project is Lead Outcome. A lead is counted as a success if it advances to Sales Qualified Lead (SQL), Sales Qualified Opportunity (SQO), or Won. Leads that do not yet have an outcome recorded are treated as still in progress and are not included in the success rate.

MasterControl will consider the project successful if our recommendations identify the account types and job titles most likely to progress for Mx, enabling Marketing and Sales to focus outreach and resources on higher-converting profiles. Success will be measured by increasing the Mx progression rate from the current 12.7% to an estimated 16–18%, closing a meaningful portion of the gap toward Qx's 19.7% benchmark.

Details:

This project will be executed by the data analytics team: Corinn, Josh, Joel, and Gaby. We will meet weekly to review progress, discuss findings, and assign next steps. The project will be completed by April 8, 2026, aligned with the course's final presentation deadline.

Milestones

- **February 1st:** Business Problem statement finalized and submitted
- **February 22nd:** Complete EDA and summarize key patterns
- **March 22nd:** Finalize model selection, and complete performance evaluation and interpretation of results
- **April 5th:** Practice presentation and refine the storyline, visuals and recommendations based on feedback
- **April 8th:** Final presentation and delivery of targeting recommendations